Science

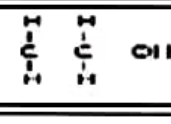
Section A — समूह 'क'

(Multiple Choice Questions) **(10×1 = 10)**

Q.No.	Correct Answer	Marks
1a	ii. Watt	1
1b	iii. A → Platyhelminthes, B → Nemathelminthes	1
1c	iii) Single-stranded structure containing uracil	1
1d	ii. Rhizome, because it contains the highest concentration of curcumin	1
1e	iii. 2.45 m/s^2	1
1f	ii. Specific heat capacity is directly proportional to change in temperature (Incorrect statement)	1
1g	iii. Statement and argument 1 are correct but argument 2 is incorrect	1
1h	ii. Closed universe	1
1i	iv. Orbitals with lower energy are filled first	1
1j	i. Primary preservatives are naturally derived…	

Section B — समूह 'ख'

(Very Short Answers) **(9×1 = 9)**

Q.No.	Correct Answer	Marks
2a	Maintains balance between technology use and health or Reduces stress, addiction, and eye strain or Improves focus, sleep, and relationships or any other ….	1
2b	Because it gives birth, breastfeeds, has lungs, warm-blooded	1
2c	It reduces the number of chromosome to half	1
2d	Water expands on freezing (anomalous expansion)	1
2e	Between F and optical centre (O)	1
2f	Gravitational force	1
2g	7 periods and 18 groups	1
2h	Calcium Oxide (CaO)	1
2i	Structural formula of ethanol: 	1

Section C — समूह 'ग'

(Short Answers) **(14×2 = 28)**

Q.No.	Correct Answer	Marks
3	s = v/t applies only to uniform motion; units mismatch Units check: m/s ÷ s = m/s², not *m*.	2
4	▯ They bear naked seeds (not enclosed in fruits). ▯ Male and female cones are produced separately. Or any two characteristics of gymnosperms	2
5	▯ A & B reproduce sexually from fertilized eggs. ▯ C reproduces from unfertilized egg (parthenogenesis) → male drones.	2
6	▯ F_1 genotype = Tt ▯ Phenotype = all tall ▯ Demonstrates Law of Dominance and its statement	2
7	High = thrombocytosis clotting risk; low = Haemophilia bleeding do not stop	2
8	▯ Avoid poaching / illegal trade ▯ Reduce plastic and pollution in natural habitats Or any two conservation actions	2
9 (i)	Pressure transmitted through liquid = 7500 Pa (P=F/A)	1
9 (ii)	$\frac{F1}{A1} = \frac{F2}{A2}$ Load lifted by large piston = 900 N	1
10	Using heat lost = heat gained: 5×(100 – 36.25) = 15×(36.25 – T)→ Initial temp = 15°C	2
11	Concave lens; diverges rays to correct myopia	2
12(i)	Increasing atomic number	1
12(ii)	Smallest atomic radius: Ar because Atomic radius decreases across a period due to increased nuclear charge	1
13	(i) Decomposition ($KClO_3 \rightarrow KCl + O_2$), requires catalyst ($MnO_2$) (ii) Combination ($Na + Cl_2 \rightarrow NaCl$), requires sunlight/heat Or any two differences (decomposition vs combination, catalyst, sunlight etc.)	2
14	▯ Concentration of ore ▯ Roasting / Calcination ▯ Reduction ▯ Purification (Refining)	2
15a	Because they are less reactive (little affinity)	1
15b	n=2 → C_2H_6 (ethane); used as fuel / LPG component or any one use	1
16	1. Excessive pesticides harm human health and cause diseases … 1	2

Q.No.	Correct Answer	Marks
	2. They damage the environment by killing beneficial insects, polluting soil and water. ....1 Or any other	

Section D समूह 'घ'

(Long answer type question लामो उत्तर आउने प्रश्नहरू. $(7 \times 4 = 28)$

Q.No	Expected Answer	Marks
17.	i. Broadband vs Baseband: Broadband—high bandwidth, multiple signals simultaneously; Baseband—low bandwidth, single signal. (2) ii. Advantages: High speed / long distance / better data handling (any one). (1) iii. Difference with analogue: Digital less affected by noise, allows better signal processing; analogue more prone to noise. (1)	4
18.	i. Name of process: IVF (In Vitro Fertilization) (1) ii. A = Sperm / male gamete, B = Ovum / female gamete (1) iii. Advantage: Helps infertile couples / controlled fertilization; Disadvantage: Expensive / technically complex (2)	4
19.	Blood pressure: Pressure exerted by blood on artery walls. (1) Symptoms: Headache, dizziness, nosebleed, blurred vision (any 3). (1+1+1)	4
20.	• i. $s_1 = \frac{1}{2}gt^2 = \frac{1}{2} \cdot 9.8 \cdot 2^2 = 19.6\,m$ (1) • $s_2 = vt - \frac{1}{2}gt^2 = 20 \cdot 2 - 19.6 = 20.4\,m$ (1) i. Height of tower: $h = s_1 + s_2 = 19.6 + 20.4 = 40m$(1) ii. Apparent weight of falling ball: $W' = 0$(freely falling) (1)	4
21.	i. Condition: Mirage (1) ii. Main cause: Total internal reflection (TIR) due to temperature gradient (1) iii. Two conditions: Hot surface creating air layers, incident light angle > critical angle for TIR (any two) (2)	4
22.	i. Use of transformer: Step-up / step-down in power transmission (1) ii. Output voltage: $V_s = \frac{N_s}{N_p}V_p = \frac{300}{1000} \times 220 = 66V$(2) iii. Core laminated to reduce eddy current losses (1)	4
23.	i. Gas collected: Ammonia (NH_3) (1) ii. Tank X: Lime tower ($Ca(OH)_2$) used to purify / absorb CO_2 or other impurities from the gas/to obtain dry ammonia gas (1) iii. Test: Red Litmus paper turns blue / pungent smell (1) iv. Agricultural use: Fertilizer preparation / soil enrichment (1)	4